

Cabbage Bacterial Leaf Spot

Recognize the problem

Small brown spots appear on leaves and spots extend or join to form irregular areas of dead tissue often limited by the leaf veins. Eventually all the leaves become yellow and rotten.

Background

The bacteria live in the affected areas of the plant and like wet conditions. They enter plants through damaged areas, even tiny amounts of damage (such as from insects chewing or sand blown against the leaves) can let bacteria in. The bacteria cannot move themselves and need to be carried to the healthy plants by insects, rain splash or weeding tools

Close up of typical leaf spots on cabbage.



Management

- Seeds should be disease-free before planting, make sure your seed is from a clean source.
- Tolerant varieties should be selected where possible.
- Try not to grow cabbages on the same land year after year even if they are healthy.
- When symptoms are seen the infected plants should be carefully removed.
- Infected plant material will provide a large amount of bacteria ready to infect the next crop so try and reduce the amount of infected material being returned to the soil.
- Mulching the crop will reduce rainwater splashes to the plants, thus reduce spread of the infection/disease.
- Avoiding weeding during damp conditions, accidental damage to the plants will allow the bacteria into the plants.
- Use bleach or fire on the weeding tools regularly so as to kill any bacteria on them.
- Chemical control is not easy for this disease however the cabbage crop may be sprayed with copper oxychloride which will protect the plants from further infection but will not control disease within the infected plants.

Bacterial leaf spot on whole cabbage.



When using a pesticide, always wear protective clothing and follow the instructions on the product label, such as dosage, timing of application, and pre-harvest interval.

Scientific name(s) > *Alternaria brassicicola*

The recommendations in this factsheet are relevant to: Ghana, Uganda



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Edited by Plantwise.

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Common pests and diseases on cabbage in Ghana



Black rot

Xanthomonas campestris pv. *Campestris*
(CABI. Teresia Kodwaran, Kenya)

- Yellow areas along leaf edges which expand to the leaf midrib, giving a distinctive 'V' shaped lesions with yellow margins and brown necrotic tissue in the centre
- Leaf veins can appear brown/black
- Plants may be stunted
- Heads are smaller and outer leaves may die



Cabbage ring spot

Mycosphaerella brassicicola (syn. *Cercospora albomaculans*)
(David B. Langston, bugwood.org)

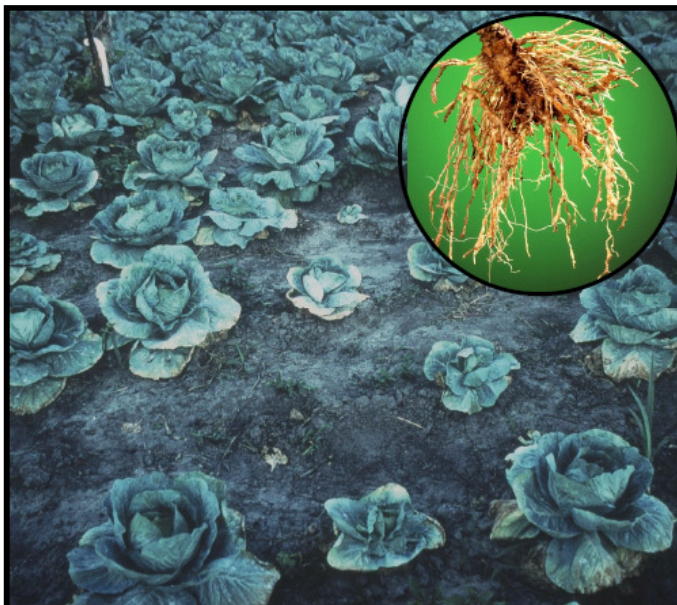
- Light brown grey/dark brown spots surrounded by a yellow halo, up to 20 mm wide angular and **limited by veins**, tiny black dots (fruiting bodies) inside the spots, arranged in concentric circles
- Spots occur on seedlings but usually on older leaves of mature plants



Leaf spot/black spot

Alternaria brassicicola
(Gerald Holmes, Bugwood.org)

- Light brown to grey/dark brown spots surrounded by a yellow halo, up to 25 mm wide, tiny black dots (fruiting bodies) inside the spots arranged in concentric circles. **Lesions are not limited by leaf veins as they are with Cabbage ring spot disease**
- Spots occur on seedlings but usually on older leaves of mature plants



Nematodes

Meloidgyne spp.
(CABI. Scott Bauer, Bugwood.org)

- Affected plants appear in patches
- Stunted and yellow, may wilt in hot weather
- Roots are severely distorted and swollen, with lumps (galls) which may be pinhead-sized to 25 mm wide



Common pests and diseases on cocoyam in Ghana



Aphids

Brevicoryne brassicae

(Kimberly Steinmann and Alton N. Sparks, Bugwood.org)

- Adults: Greyish-green body with greyish-white to powdery blue waxy covering. Found in clusters on lower leaves, stems or growing points
- Nymphs: brownish, on underside of leaves and stems
- Wilting or yellowing of plants and the presence of sooty mould where aphids have been feeding
- Ants may be protecting the aphids



Diamondback moth (DBM)

Plutella xylostella

(Sipher Wu Bettaman, David Coppert, Bugwood.org. A. M. Varela, CABI)

- Small, grey moths which have a diamond pattern on their wings when resting
- Caterpillars are yellowish/green, around 1 cm long when fully grown. Hang from a web when disturbed and often appear at the 4 leaf stage
- Leaves: irregular holes, windowing and skeletonised underside

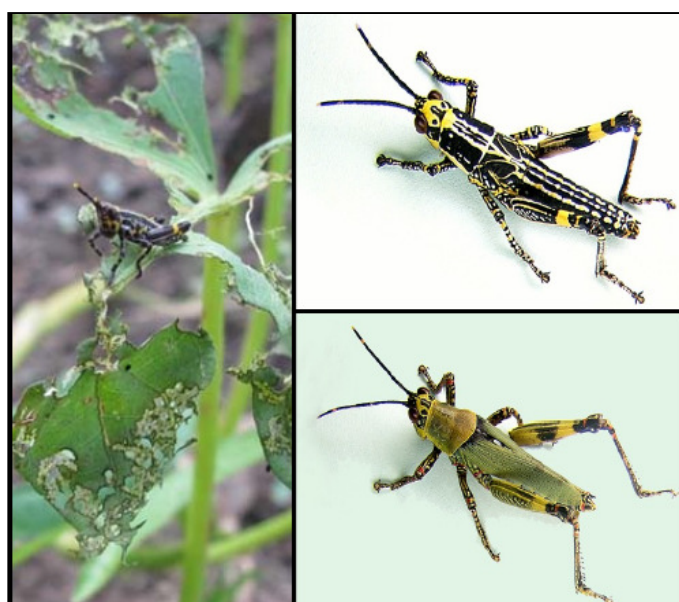


Webworm

Hellula undalis

(Alton N. Sparks and Natasha Wright, Bugwood.org)

- Larvae: creamy white, light pinkish-brown stripes along the body, Dark brown/black head. Feed on underside of leaves. Up to 12-15 mm long. Often covered in webs which become covered in dirt and frass.
- Adult: Sandy brown moth with a black spot and zigzagging light brown lines on wings, about 12 mm wide
- Plant stunting, multiple growing points/heads, mining of leaves, bore holes in stems



Grasshopper

Zonocerus variegatus

(Georg Goergen, IITA Insect Museum)

- Juveniles: 2mm long at their smallest. All instars have yellow and black stripes parallel to the body axis
- Adults: can grow to 30-52 mm, yellow and black stripes with red spots, main body is a yellow-green colour
- Chewed leaves and stems

A close-up photograph of a cabbage plant. The leaves are large, green, and show significant signs of damage, including large holes and irregular, light-colored necrotic patches, likely caused by pests or diseases.

CABI PEST AND DISEASE PHOTOGUIDE TO

Cabbage disorders

KNOWLEDGE FOR LIFE

Introduction

This photo booklet has been produced by the CABI-led **Plantwise** programme (www.plantwise.org) to aid extension officers and other plant health advisors in diagnosing the most common pests, diseases and abiotic problems of cabbage around the world. The symptoms presented on a real plant sample can be compared with the photos in this guide to identify possible causes.

The booklet is organized into two broad sections, one showing the common insect pests that attack the crop and the other showing the various symptoms of poor health. In the symptoms section, the images are arranged by plant part, with similar-looking symptoms displayed together. Some biotic and abiotic factors cause more than one type of symptom, so there may be multiple images in different parts of the photo booklet for a specific problem. The photos for a particular problem are cross-referenced to make it easy to find all the relevant photos.

Contents

Sign or symptom	Box #
Insects	1–34
Leaf	35–50
Stems	51–59
Whole Plant	60–68
Root	69–70

Turnip moth

Agrotis segetum



Photo: F. L. Martin GNU licence

- The larvae are generally grey sometimes looking a little purple.
- Attacks the lower stems often results in cutting off of seedlings.
- Often found curled up near the base of plants.

Cabbage leaf sawfly

Athalia proxima



Photo: Merle Shepard, Gerald R. Carner, and P.A.C. Ooi, Insects and their Natural Enemies Associated with Vegetables and Soybean in Southeast Asia, Bugwood.org

- Milk white eggs 0.6-0.9 mm high.
- The larva is NOT a true caterpillar (there are more than 4 false feet on the body of the larvae).
- Early instars are grey/dark grey, as they age the head turns black.
- Tough silken yellow cocoon.

Cabbage looper

Trichoplusia ni



Photo: Distant Hill Gardens, Flickr

- Initially larvae are greyish/white, later pale green and marked with distinct white stripe on each side.
- Larvae have three pairs of prolegs; crawl by arching back to form a loop and then projecting the body forward.
- Eggs yellowish, white or greenish, 0.6 mm diameter and 0.4 mm tall, laid singly or in small clusters.

Black cutworm

Diaphorina citri



Photo: B Smith, Flickr

- Larvae are often same colour on back and underneath, ranging from light grey/grey-brown to nearly black; underside may be lighter.
- Initially feed on leaves but older caterpillars will cut the plant off at ground level.
- Mainly a pest of young plants as older plants are more resistant.

Cabbage webworm

Hellula rogatalis



Photo: A. N. Sparks Jr. University of Georgia, Bugwood.org

- Older larvae have 5 dark stripes from head to tail and yellowish or brownish hairs.
- Larvae produce a lot of silk, for protection.
- Eggs laid singly or in small masses on the youngest leaves, oval and flattened in shape, greeny grey.

Egyptian cottonworm

Spodoptera littoralis



Photo: Biologische Bundesanstalt für Land- und Forstwirtschaft, Biologische Bundesanstalt für Land- und Forstwirtschaft, Bugwood.org; O. Heikinheimo, Bugwood.org

- Hairless larvae, 40–45 mm long, dark and light bands running down sides with black patches at both ends.
- Eggs laid in regular rows, 0.6 mm in diameter. White initially but black before hatching, often in mass of hair scales (see insert).

Cotton leafworm, Tobacco cutworm

Spodoptera litura



Photo: Yuan-Min Shen, Taichung District Agricultural Research and Extension Station, Bugwood.org

- Young larvae pale green to dark green, older larvae are brown.
- Yellow stripes along the back and the sides, with a row of black dots along each side and a row of dark triangles each side of the central line.

Diamondback moth

Plutella xylostella



Photo: A. N. Sparks Jr. University of Georgia, Bugwood.org

- Full-grown larvae are 10mm long; body green but can be yellowish with distinct body segments, and bears a few short hairs.
- Head pale to pale greenish or pale brown, mottled with brownish and black-brown spots; eye spot is black.

Cabbage cluster caterpillar

Crocidolomia pavonana



Photos: Merle Shepard, Gerald R. Carner, and P.A.C. Ooi, Insects and their Natural Enemies Associated with Vegetables and Soybean in Southeast Asia, Bugwood.org

- The later larval stages have whitish stripes, 3 on the back and one on each side.
- Eggs are laid in masses of up to 300 at a time and held together with a glue (see insert right), greenish in colour, becoming red-brown later.
- Pupae are yellowish green when formed but turn dark brown later.

Large cabbage white butterfly

Pieris brassicae



Photos: P. Taylor, CABI

- Larvae up to 30 mm long; yellow and black with obvious hairs.
- Often initially on the underside of leaves or hidden among young leaves.
- They produce large amounts of black frass (excrement).

Small cabbage white butterfly

Pieris rapae



Photo: S. Nelson, Flickr

- Larvae up to 25 mm long; green or bluish-green with thin yellow line on back.
- Body covered with short fine hair.

Cabbage root fly

Delia radicum



Photos: Rasbak, Wikimedia.org

- Larvae (up to 6 mm) white and pointed at one end; burrow into roots (main picture).
- Flies look like small house flies, 6 mm long (insert left).
- The fly lays white, elongated eggs at the base of plant.
- Pupae are brown and are also found in the soil (insert right).

Egyptian cottonworm

Spodoptera littoralis



Photo: O. Heikinheimo, Bugwood.org

- Grey brown body, wingspan 30-38 mm.
- Forewings grey to reddish brown.
- Rear wings greyish white with grey edges.

Cabbage webworm

Hellula rogatalis



Photo: Natasha Wright, Cook's Pest Control, Bugwood.org

- Yellowish brown front wings with bands and a dark kidney shaped spot; greyish rear wings.
- Body 20 mm long, wingspan 13 mm.
- Resting on the ground where it is well camouflaged the moth takes short random flights when disturbed.

Cotton leafworm, Tobacco cutworm

Spodoptera litura



Photo: D. Hobern, Flickr

- Adult moths 15-20 mm, with wingspan of 30-38 mm.
- Forewings grey to reddish-brown, with a complex pattern of creamy streaks and paler lines along the veins. Rear wings are greyish-white with greyish-brown margins.

Diamondback moth

Plutella xylostella



Photo: IITA, Flickr

- Adult small, slim, greyish-brown moth with pronounced feelers (antennae).
- Marked with a broad cream or light brown band along the back.
- When viewed from the side, the tips of the wings can be seen to turn upward slightly.

Cabbage cluster caterpillar

Crocidolomia pavonana



Photo: Goldentakin, Flickr

- Wingspan about 30 mm; black thorax and a reddish brown body.
- They emerge at night.
- Larvae initially feed on the underside of the leaves but then eat the whole leaves and finally the stem.

Large cabbage white butterfly

Pieris brassicae



Photo: Hania Berdys, Bugwood.org

- Wingspan 55–70 mm, with upper sides usually very white, with a pronounced black tip to the forewing.
- The head, and body are black with grey hair-like scales.

Turnip moth

Agrotis segetum



Photo: T. Morris, Flickr

- Shows various colour forms, although usually the forewings show a neat, pale fringe with a narrow inner dark line and the rear wings are white.
- Wingspan is 32–42 mm.
- Flies at night.

Small cabbage white butterfly

Pieris rapae



Photo: Forrest and Kim Starr, Starr Environmental, Bugwood.org

- Wingspan 32–47 mm; upper side is creamy white with black tips on the forewings. Females also have two black spots in the centre of the forewings. Underwings are yellowish with black speckles.
- Looks like a smaller version of the Large cabbage white butterfly (*Pieris brassicae*).

Black cutworm

Agrotis ipsilon



Photo: Whitney Cranshaw, Colorado State University, Bugwood.org

- Forewings have small, black dagger-like markings extending towards wing tip and a whitish band extending across the wing; wingspan 40-55 mm.
- Eggs about 0.5 mm in diameter, ball shaped, with a slightly flattened base; laid in clusters on leaves.

Cabbage looper

Trichoplusia ni



Photo: Joseph Berger, Bugwood.org

- Forewings are mottled grey-brown in colour with silvery white spots centrally, with a U-shaped mark and a circle or dot that are often connected (which is characteristic of this pest).
- Rear wings light brown at the base with ends dark brown. Wingspan 33-38 mm.

Large cabbage white butterfly

Pieris brassicae

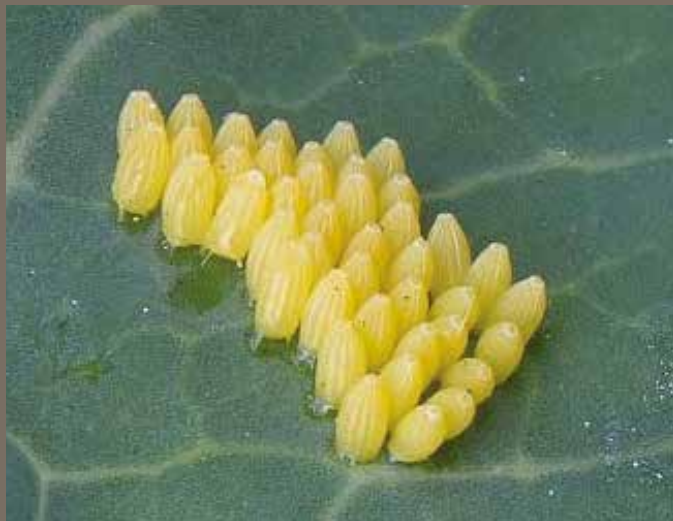


Photo: Rasbak Wikimedia.org

- Eggs bright yellow, bottle-shaped, 1.4 mm high, ribbed vertically and laid upright in clusters of 40-100.

Small cabbage white butterfly

Pieris rapae



Photos: Phil Sloderbeck, Kansas State University, Bugwood.org Merle Shepard, Gerald R. Carner, and P.A.C Ooi, Insects and their Natural Enemies Associated with Vegetables and Soybean in Southeast Asia, Bugwood.org

- Yellowish to cream coloured, bullet shaped, deeply ridged eggs laid on the leaves of host plants.
- Often laid singly or in few numbers on the leaves of both wild and cultivated brassicas.

Swede midge

Contarinia nasturtii

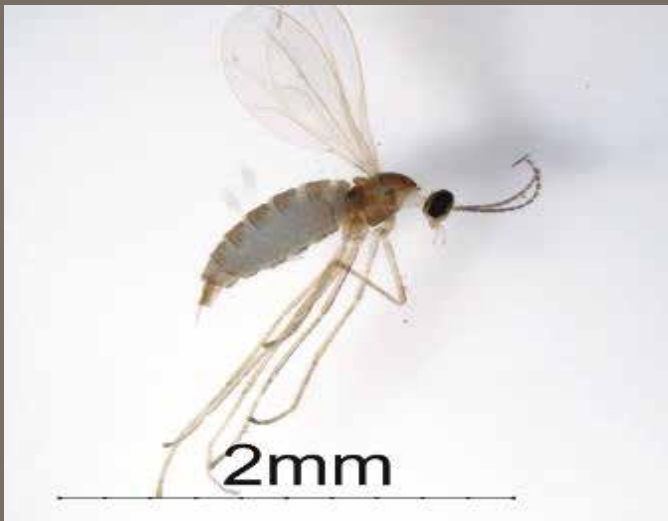


Photo: Susan Ellis, USDA APHIS PPQ, Bugwood.org

- A tiny, light brown fly that looks a bit like a mosquito.
- Several generations a year, which means adults can be seen in any month.
- The pupa is found in the soil.

Cabbage leaf sawfly

Athalia rosae



Photo: D. Hobern, Flickr

- Orange middle section with black markings; abdomen yellow/orange with black ovipositor.
- Clear membranous wings with obvious venation.

Vegetable leaf miner

Liriomyza sativae



Photo: Central Science Lab, Bugwood.org

- Tiny yellow and black fly.
- Middle section is shiny black.
- Not active fliers and will only take short jerky flights from one leaf to the next; often seen walking around on leaf surfaces.

Cabbage whitefly

Aleyrodes proletella



Photo: Rasbak, Wikimedia.org

- Adults 1.5 mm long; white wings with four grey spots – wings white due to a powdery wax. Head dark with yellow underside; red eyes.
- Often seen on leaf underside, flying only when disturbed.
- Nymphs are yellow and scale-like.
- Can transmit viruses.

Harlequin bug

Murgantia histrionica



Photos: Whitney Cranshaw, Colorado State University, Bugwood.org

- Black, shield-shaped stinkbug, brilliantly marked with red, yellow and orange.
- Young (nymphs) appear similar to adults.
- Feeding damage is yellow or white slivery areas on leaves and heavy infestations can cause plants to wilt, turn brown and die.
- Eggs are barrel-shaped and laid together.

Mealy cabbage aphid

Brevicoryne brassicae



Photo: E. Finkle, Flickr

- Very closely packed colonies of grey waxy aphids, each aphid about 2.5 mm long; mostly without wings.
- Often found on undersides of leaves.
- Leaves can become distorted and reddened, even when infestations are small.
- Can transmit viruses.

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Onion thrips

Thrips tabaci



Photos: Whitney Cranshaw, Colorado State University, Bugwood.org;
A. N. Sparks Jr University of Georgia, Bugwood.org

- Young (nymphs) and adults very small 0.5–1.2 mm. The body is thin, and slender. Their eyes are darker and are easy to see.
- Often like to crawl into regions between leaves.

Green peach aphid (Peach potato aphid)

Myzus persicae



Photo: C. Quintin, Flickr

- Can be many colours from yellow green or pink but any one aphid is the same colour all over.
- Winged forms are usually black.
- Usually found on lower sides of leaves in sparse colonies.
- Transmits a lot of viruses.

Mustard aphid

Lipaphis erysimi



Photo: A. N. Sparks Jr University of Georgia, Bugwood.org

- Olive-green to brown, with ridges on the back.
- Found on leaf underside, young shoots, growing points, and when severe on both sides of the leaf.
- Causes leaf rolling, yellowing and stunting.
- Severe infestations prevent head formation and flowering.

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Crucifer flea beetle

Phyllotreta cruciferae



Photo: Whitney Cranshaw, Colorado State University, Bugwood.org

- Small, oval-shaped, blackish beetle.
- Adults will jump into the air like a flea if you get close to them.
- Larvae and adults attack seedlings. They produce holes in the seedlings leaves and can distort them.
- Do not damage older plants.

Black spot Grey leaf spot Target spot

Alternaria spp. commonly *Alternaria brassicicola*



Photo: Gerald Holmes, California Polytechnic State University at San Luis Obispo, Bugwood.org

- Seen in the seedbed as leaf spots and damping off.
- Small dark brown/black leaf spots, which enlarge and have a target ring pattern.
- Dusty black (spores) material may be seen on fingers after touching the spots.

Black spot Grey leaf spot Target spot

Alternaria spp.



Photo: Gerald Holmes, California Polytechnic State University at San Luis Obispo, Bugwood.org

- Centres of the leafspots split and drop out.

Black Rot

Xanthomonas campestris



Photos: J. Mulema, CABI

- Very common disease.
- Characteristic V-shaped lesions spread in from the edge of older leaves or from insect damage.
- The spreading edge of the lesion is yellow whereas the older areas are brown.

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Calcium deficiency

Calcium deficiency

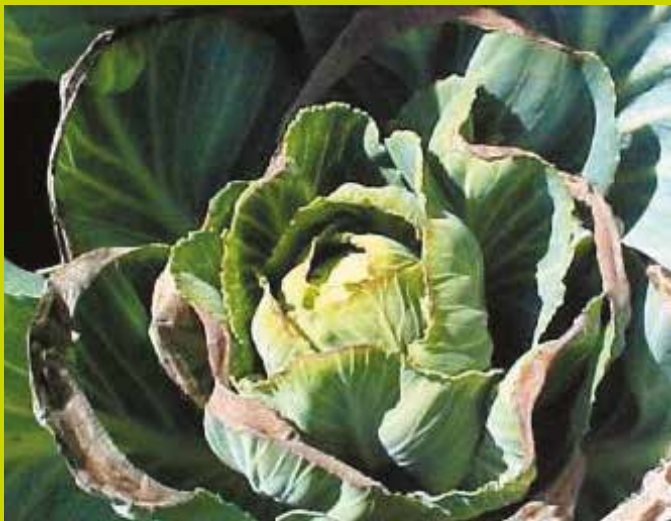


Photo: David B. Langston, University of Georgia, Bugwood.org

- Edges of leaves turn brown.
- Can be hidden inside the cabbage head on enclosed leaves and so not visible at harvest.
- Necrosis does not spread to neighbouring leaves in the same way a pathogen does.

Leaf miner

Liriomyza spp.



Photo: P. Taylor, CABI

- Long winding tracks within the leaf.
- Hold up the leaf up to the light to see the larva and frass (excrement) within the leaf mine.

Onion thrips

Thrips tabaci



Photo: Gerald Holmes, California Polytechnic State University at San Luis Obispo, Bugwood.org

- Silvering of leaves which may become brown and hard (corky) as the plant responds to the damage (by this time the thrips may have moved on).
- Thrips are often associated with oily droplets of frass (excrement) deposited on the leaf surface (seen with a handlens).

Downy Mildew

Hyaloperonospora brassicae



Photo: Gerald Holmes, California Polytechnic State University at San Luis Obispo, Bugwood.org

- Usually a leaf disease but can attack seedling leaves, stems of seedlings and the white region of cauliflower.
- Can be vein limited in some cases.
- In cool weather there may be little yellowing (initially) but with time or in hot dry weather the plant material will become yellow and then brown.

Ring spot

Mycosphaerella brassicicola



Photo: M. Gammelgaard, platesygdomme.dk

- Well defined spots 10-20 mm, usually tan coloured and covered in black dots (fruiting bodies).
- The border of the spots is grey/olive colour and this may remain after the rest of the leaf has yellowed.

Soft rot of cabbage / Bacterial black Stalk

Pectobacterium carotovorum ssp. *carotovorum*



Photos: P. Taylor, CABI; Gerald Holmes, California Polytechnic State University at San Luis Obispo, Bugwood.org

- Lower end of the mid-rib rots and loses structure.
- A characteristic extremely unpleasant smell.
- Spreads along leaf veins.
- Bacterial ooze can be seen leaking from the rotting area.

White leaf spot

Mycosphaerella brassicicola / *Pseudocercospora capsellae*



Photos: David B. Langston, University of Georgia, Bugwood.org

- Appears as tan or light brown spots, they may join together and may drop out.
- Can appear similar to Paraquat damage.
- Stem lesions are elongated and brown at first, turning ashy grey to white with a brownish margin.

Powdery Mildew

Erysiphe cruciferarum



Photo: J. Mulema, CABI

- White powdery growth that spreads over the upper surface of leaves.
- Initially the leaf will appear healthy beneath but will become brown with time.
- Characteristic mushroom-like smell.

White rust

Albugo candida



Photo: P. Taylor, CABI

- Various amounts of browning can occur; green islands can remain after the leaf has yellowed.
- Often associated with blistering and distortion of the leaves or shoot.
- Can be seen on all parts of the plant.

Edema



Photo: David B. Langston, University of Georgia, Bugwood.org

- Swellings, usually on the underside of leaves; some cultivars more susceptible than others.
- Non-pathogenic disorder that will not spread from one plant to another.
- It is caused by a combination of high humidity and high soil moisture content, might also be the result of some chemical sprays.

Ennations



Photo: P. Taylor, CABI

- Small leaflets appear on the main leaf laminar.
- Unusual symptom for which the cause is not known.

Downy Mildew

Hyaloperonospora brassicae



Photo: Gerald Holmes, California Polytechnic State University at San Luis Obispo, Bugwood.org

- Downy growth usually on the underside of leaves but can be on both sides if severe.
- This may be sparse (as shown above) or very downy and like fluff.

Whiptail of crucifers

Molybdenum deficiency



Photo: P. Taylor, CABI

- Leaves develop a boat-like appearance if the plant is short of molybdenum.
- If very severe the leaf edges are so reduced that the mid rib is all that remains, with the leaf appearing as though it's been eaten.

Root rot damping off

Rhizoctonia spp. and others



Photo: Gerald Holmes, California Polytechnic State University at San Luis Obispo, Bugwood.org

- May prevent emergence of the seedling.
- If they emerge the base of the young stem is partially or totally destroyed by brown or black rot.
- They generally will fall over due to the rot and die.

White mould

Sclerotinia sclerotiorum



Photo: Gerald Holmes, California Polytechnic State University at San Luis Obispo, Bugwood.org

- Usually starts where leaves are in contact with the ground and spreads up the plant leaf by leaf.
- The head (sometimes the roots) eventually invaded.
- Many hard black fungal balls (sclerotia) can be found among the rotting leaves along with the fluffy white mycelium.
- Can develop in store.

Cabbage root fly

Delia radicum



Photos: Clemson University – USDA Cooperative Extension Slide Series, Bugwood.org
Mary Ann Hansen, Virginia Polytechnic Institute and State University, Bugwood.org

- Stunted plants that wilt easily.
- Root system very poorly developed and plants easily lifted from the soil.

Swede gall midge

Contarinia nasturtii



Photo: Julie Kikkert, Cornell Cooperative Extension, Bugwood.org

- Swollen leaf stalks and the deformed leaves.

Boron deficiency



Photo: Gerald Holmes, California Polytechnic State University at San Luis Obispo, Bugwood.org

- Symptoms are generally seen in the young leaves first. They don't expand properly and snap easily.
- Stems are hollowed out.
- Occurs most frequently on coarse or sandy acid soils. Boron becomes less available during drought.
- Easily confused with Calcium deficiency.

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Bacterial black stalk

Pectobacterium carotovorum ssp. *carotovorum*



Photo: David B. Langston, University of Georgia, Bugwood.org

- Vascular browning; wilting, darkening and water soaked stems and leaves.
- Total plant collapse.
- Very similar to a severe infection by *Xanthomonas* in the stem.

Heart rot

Calcium deficiency



Photo: S. Nelson, Flickr

- “Decay” only seen in the central stem and does not spread to leaf stalks.
- Can be mild with a small cavity and browning or can be severe and appear similar to bacterial infection.
- No bad smell is associated with heart rot.

Phytophthora root and stem rot

Phytophthora drechsler



Photo: Paul Bachi, University of Kentucky Research and Education Center, Bugwood.org

- Internal rotting of the stem often with few external symptoms.
- Unlike many other common *Phytophthoras* this attacks the roots and the middle of the stem.

Black rot of cabbage

Xanthomonas campestris

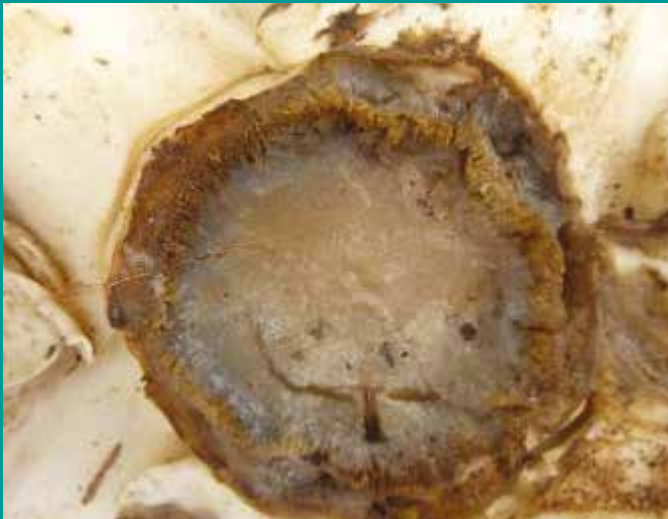


Photo: P. Taylor, CABI

- Rotting of the stem on this scale will only really occur after a lot of damage to the upper leaves.

Club root

Plasmodiophora brassicae



Photo: Gerald Holmes, California Polytechnic State University at San Luis Obispo, Bugwood.org

- Plants wilt readily.
- Often very stunted and never put on much growth depending on how badly the plants are affected.

Fusarium wilt

Fusarium oxysporum f.sp. conglutinans



Photo: Ontario MoA, Food and rural affairs

- Yellowing of the lower leaves, often initially on one side of the plant they will later turn brown and drop off.
- Brown discolouration inside the stem is characteristic.
- Eventually the entire plant may yellow, wilt, and collapse.

Turnip mosaic virus

TuMV



Photo: Horticultural Research International

- Symptoms vary with host and environment.
- Mosaic and or browning is common. Tips of leaves bending out and down is common.
- Necrotic spotting (black necrotic spots and ringspots) may develop in field or in store.
- Aphid transmitted but with limited direct plant to plant spread.

Cabbage leaf curl virus

CaLCuV

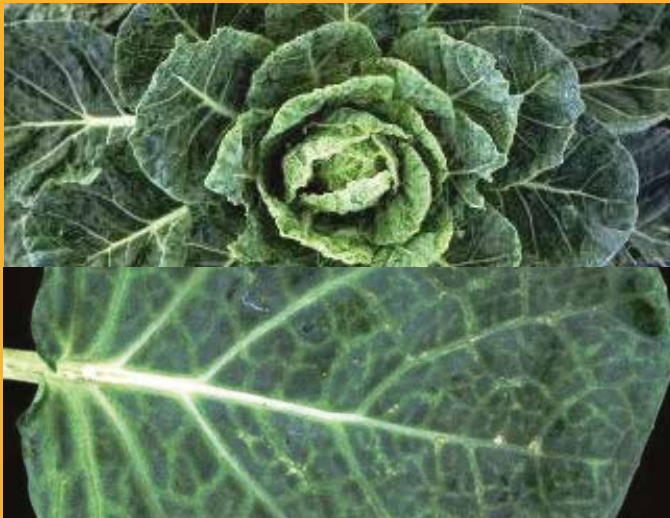


Photo: David B. Langston, University of Georgia, Bugwood.org

- Symptoms on leaves are yellow spots, vein clearing, mosaic, curling, and wrinkling.
- Transmitted by whitefly.

Cauliflower mosaic virus

Ca MV



Photos: Horticultural Research International

- Vein clearing and mild leaf distortion.
- Transmitted by aphids.
- If the weather is warm the symptoms are not seen.

Sugar beet cyst nematode

Heterodera schachtii



Photo: DAFF Archive, Bugwood.org

- General stunting of the plant.

Herbicide damage

Clomazone



Photo: Gerald Holmes, California Polytechnic State University at San Luis Obispo, Bugwood.org

- It is not possible to include all the symptoms that herbicides can induce; bleaching and leaf distortion are often typical.
- Each herbicide chemical may produce characteristic symptoms but a conversation with the farmer may be the quickest means of diagnosis.

Herbicide damage

Imazapic



Photo: David B. Langston, University of Georgia, Bugwood.org

- It is not possible to include all the symptoms that herbicides can induce; stunting and purple leaves can occur (as above).
- Each herbicide chemical may produce characteristic symptoms but a conversation with the farmer may be the quickest means of diagnosis.

Bird damage

(Pigeons)



Photo: P. Taylor, CABI

- May appear similar to locust or caterpillar damage.
- Leaves are torn rather than nibbled.
- No frass is seen.

Sugar beet cyst nematode

Heterodera schachtii



Photo: Gerald Holmes, California Polytechnic State University at San Luis Obispo, Bugwood.org

- Tiny yellow cysts seen on the roots, considerably smaller than nematode galls and unlike the galls the cysts are all the same size.

Club root

Plasmodiophora brassicae



Photo: Gerald Holmes, California Polytechnic State University at San Luis Obispo, Bugwood.org

- Above ground the plants are stunted and wilt readily, often have a blue tinge.
- Below ground large bulbous swellings on the roots are extremely characteristic.
- Initially these are white but they darken with age and may start to decay.

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Black rot on cabbage

Xanthomonas campestris pv. *campestris*



Symptoms on cabbage
(Florence Wakhungu)



Cabbage black rot (Moses Cheruiyot)



Black rot of cabbage (Gathogo GN, Kenya)

Prevention

- Use certified seed and seedlings free from disease as this disease can be seed-borne
- Use resistant varieties where available
- Do not plant crucifers in areas: that have a history of the disease; where crucifers have been grown within the previous 2 years; where cruciferous weeds are numerous
- Deep plough fields to bury contaminated crop residues
- Sterilise the seedbed by heaping and slowly burning dry grass or other plant trash on the seedbed.
- Plant on raised beds for better drainage. Avoid waterlogging.
- Do not damage leaves of seedlings as this makes them prone to disease
- Correct plant spacing: 50 cm between plants and 60 cm between rows. If plants are closer together this can encourage spread of disease between plants
- Avoid flood/furrow irrigation because it will carry the disease from infected to uninfected areas. Also avoid irrigating early in the day when dew is present or late in the day.
- Destroy alternative host weeds in the crucifer family e.g. birdsrape mustard, Indian mustard, black mustard
- Avoid moving through fields when the plant are wet to prevent disease spread. Clean tools and boots.
- Rotate crucifers for a minimum of 3 years with crops that are not susceptible such as maize, legumes, tomato

Monitoring

- Additional relevant crops: cauliflower, chinese cabbage, brussel sprouts, turnip, mustard, broccoli, rape, radish
- Monitor plants weekly for symptoms:
- Look for yellow V-shaped lesions with an open part of 'V' along the edge of the leaf
- The diseased leaf area becomes brown and brittle
- Affected leaf veins turn dark brown to black
- The V-shaped lesions enlarge and the entire leaf eventually turns yellow or wilts and falls from the plant
- Infected seedlings remain stunted and become unproductive
- There is a ring of black discoloured tissue within the stem when cut
- If 6-8 plants/ha show symptoms take action
- The disease thrives in wet conditions with high temperatures

Direct Control

- Remove all infected young plants and debris from the field and destroy by: deep burying; composting well outside the field; feeding crop to livestock
- When older plants become infected, harvest as soon as possible.

Note: Pesticides may be available to control this pest. Please check with the Ministry of Agriculture in your country to find out which pesticides are registered in your country and the local restrictions for their use.

Control of Black Rot in Cabbage

Recognize the problem

The disease is caused by bacteria and can be identified on the leaf as a "V" shaped yellow colour, moving from the edge inwards and downwards into the stem. This symptom distinguishes Black rot from Fusarium wilt where symptoms move upwards from the soil level towards the stem. As the disease progresses, the yellowing part of the leaf turns brown and the tissues die. Leaf veins turn black at the final stage of the disease.

Black rot on cabbage. Note the V shaped lesions spreading from the edges of the leaves. (Photo by Dilly-Ann Bartholomew)



Background

Black rot is one of the most serious diseases of cabbage in Grenada. It poses a big problem in areas of high humidity, reducing crop yields by as much as 75-90%. The bacteria that cause Black Rot can survive and spread via wild hosts, soil, water droplets or infected seed.

Management

- Avoid growing very susceptible varieties in the rainy season. There are no resistant varieties in Grenada.
- Do not plant cabbage in a field that has had cabbage, broccoli, cauliflower, kale or any other member of the Brassica family for the previous 3 years
- Avoid using overhead sprinklers because they can cause soil to splash on leaves and so will spread the disease
- Control weeds in and around the production area, especially weeds from the cabbage family
- Do not enter fields when they are wet to reduce the risk of transferring the disease from one leaf to another by moisture on the leaves
- Prune off all leaves close to the ground and leaves that show signs of the disease (use clean tools regularly dipped in bleach)
- Bury or burn all crop debris after harvest.

Advanced Black Rot in cabbage. (Photo by Dilly-Ann Bartholomew)



Chemical Control

- Spray Copper Hydroxide (e.g. Kocide 101, Champion, or Coback) on the entire plant, paying attention to the underside of the leaves, at a rate of 2-4 Tbsp per U.S. Gallon.

When using a pesticide, always wear protective clothing and follow the instructions on the product label, such as dosage, timing of application, and pre-harvest interval.

Scientific name(s) > *Xanthomonas campestris* pv. *campestris*

The recommendations in this factsheet are relevant to: Grenada

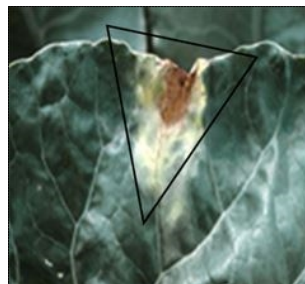


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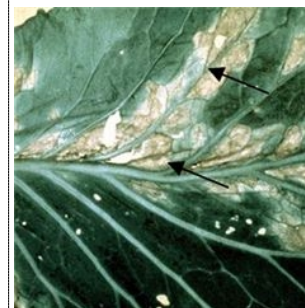
Edited by Plantwise.

Xanthomonas bacterial black rot on cabbage

Xanthomonas campestris pv. *campestris*



Cabbage leaf affected by black rot showing the V-shape (photo M.T.McGrath)



Cabbage leaf affected by black rot showing the black veins (photo ISU)

Prevention	Monitoring	Direct Control	Direct Control	Restrictions
- Use certified tolerant varieties like Riana F1, KK cross F1 and Gloria F1 - Plant on raised beds for better drainage - Sterilise the seedbed by heaping and slowly burning dry grass or other plant thrash on the seedbed. - Increase spacing in the hot season, such as up to more than 60 x 60 cm. - This bacterial disease is mainly spread by water. Therefore, do not flood irrigate from infected towards un-infected field areas. - Avoid water logging - Avoid working in the field when plants are wet as the disease can spread. - Rotate with non-brassicac like maize and legumes for at least 2 years after you had black rot disease in your field, and avoid susceptible crops like rape, Chinese cabbage, cauliflower and broccoli.	- This is a bacterial disease, common during wet conditions with high temperatures - Monitor for: Yellow, brown or black v-shaped patches at the edge of the leaves. If the yellow/black lesion is held against the light, black veins are visible. A cross section of infected stem reveals a black ring. In later stages, cabbage head turns black with an offensive foul smell - Take immediate action when first symptoms appear on young plants. - When symptoms appear at late crop stage or close to harvest, often no action is needed, because outer damaged cabbage leaves can be removed.	- Remove all infected young plants and debris from the field and destroy by deep burying or well-composting outside field. - When older plants become infected, harvest as soon as possible.	- There are no really effective pesticides against bacteria. - Only copper based products can help to prevent disease spread. - When using a pesticide or botanical, always wear protective clothing and follow the instructions on the product label, such as dosage, timing of application, pre-harvest interval, max number of sprays, restricted re-entry interval. Do not empty into drains and water sources. - WHO toxicity class II pesticides might not be allowed in local IPM schemes. - Always consult recent list of registered pesticides of ZEMA.	WHO toxicity class II (moderately acute hazardous); Pre-harvest interval p.h.i. about 2 weeks and restricted re-entry interval 1 day, but check product label. Use a maximum of 3 sprays. Spray interval of 14 days during wet weather. When used too much, copper(II)-ions can accumulate in soil and water. It can burn leaves. Livestock should be kept out of treated area for this period

Zambia

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Downey Mildew ya Njegere

Peronospora viciae f. sp *pisi* Downy mildew on Peas (English)

	Kuzuia	Ufuatiliaji	Udhibiti wa moja kwa moja	Udhibiti wa moja kwa moja	Vizuizi
 Downy mildew kwenye majani ya njegere (Chanzo: Phil Taylor, CABI)	● Lima kwa mzunguko na mimea ya mikunde ambayo sio mwenyeji wa ugonjwa. ● Panda mbegu safi zilizothibitishwa ● Ondoa sehemu za mmea zilizoambukizwa msimu uliopita na uziharibu. ● Tumia mbolea ya inayotoa madini polepole kama samadi ili kuzuia nitrojeni kupita kiasi kwani sehemu za mimea yenye majani machanga hushambuliwa. ● Epuka kutumia umwagiliaji maji wa juu ili kupunguza ugonjwa kuenea ● Simika njegere ili uongeze mzunguko wa hewa ● Punguza unyevu katika green house	● Kagua mimea ya njegere iliyodumaa, majani ya manjano na mimea iliyokauka. ● Kagua mkusanyiko wa ukungu mweupe kama pamba hadi hudhurungi na madoa ya manjano kwenye sehemu za juu na chini za majani. ● Chukua hatua wakati mmea 1 imeathiriwa	● Ondoa na uharibu sehemu za mimea zilizoambukizwa ili kupunguza kuenea kwa magonjwa kwa mimea yenye afya ● Nyunyiza bidhaa zenye mchanganyiko wa <i>Trichoderma</i>	● Tumia bidhaa zilizosajiliwa na PCPB kwa mazao lengwa-www.pcpb.go.ke ● Fuata maagizo kwenye kibandiko ili kujua muda wa kuingia (REI) ama kuvuna (PHI) ● Vaa vifaa vya kujikinga ● Usimwage mabaki ya dawa kwenye mitaro ya maji ● Nyunyiza bidhaa zenye mchanganyiko wa azoxystrobin ● Nyunyiza bidhaa zenye mchanganyiko wa Difenoconazole	● Shirika la afya duniani (WHO) kitengo cha U (Bidhaa hizi hazina uwezekano wa kusababisha madhara). Mstari wa rangi ya kijani ● Shirika la afya duniani (WHO) kitengo cha II (madhara ya kadri). Mstari wa rangi ya manjano
 Kagua mimea ya njegere iliyodumaa, majani ya manjano na mimea iliyokauka (Chanzo: Phil Taylor, CABI)					
 Downy mildew kwenye majani ya njegere (Chanzo: Phil Taylor, CABI)					

Kenya

CREATED/UPDATED: Septemba 2023

Waandishi: Nicholas Muendo (PCPB), Joseph Karani (MESPT), Miriam Otipa (KALRO), Kadenge Lewa (KALRO), Finyange Pole (KALRO), Stella Mwashumbe (KALRO), Ngombo Katana (KALRO), Miriam Otipa, Kadenge Lewa (KALRO), Finyange Pole (KALRO), Stella Mwashumbe (KALRO), Ngombo Katana (KALRO)

Hii ilehariwe na: Plantwise

Downy mildew on Peas - Kenya

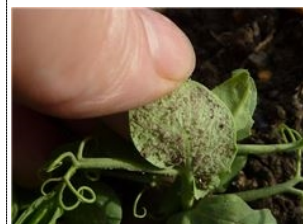
Peronospora viciae f. sp. pisi



Typical downy mildew sporulation on the underside of pea stipules. (Source: Phil Taylor, CABI)



Systemic infection by downy mildew on pea. (Source: Phil Taylor, CABI)



Typical downy mildew sporulation on the underside of pea stipules. (Source: Phil Taylor, CABI)

Prevention	Monitoring	Direct Control	Direct Control	Restrictions
- Employ crop rotation, avoid peas and faba beans for 3 years (all other crops are resistant). It is seed-borne so plant certified seeds. - In the case of severe infection, remove crop trash after harvest. - Seedlings are susceptible but plants become resistant with age. - Use a slow-release organic fertilizer such as well-decomposed manure to avoid excess nitrogen as young leafy plant parts are susceptible. - Where possible, attempt to keep leaves dry for as long as possible. - Avoid sprinkler irrigation to reduce the level of spread. - Staking peas to increase air circulation. - Reduce humidity in greenhouses.	- Look out for waxy pale yellow plants with thickened leaves. This indicates systemic infection. - Look for cottony-white to tan fungal colonies and yellowed blotches on the upper and lower surfaces of the leaves. - Take action when if you see one systemically infected plant.	- If a systemically infected plant is seen, remove and destroy it to reduce the spread of disease to healthy plants. - Plants will generally grow away from the infection because they become resistant with age. If severe, consider sprays. - Spray plants using <i>Trichoderma</i>-based products.	When using a pesticide always wear appropriate personal protective equipment.	
			Follow the instructions on the product label on dosage, maximum applications, PHI, Re-entry interval, and disposal of pesticide waste amongst others.	
			Use list of registered pesticides (PCPB): www.pcpb.go.ke	
			Try to apply a blend of fungicides to avoid the development of resistance. Spray just before petal fall especially if the weather is wet.	These restrictions apply to one active ingredient only.
			Use azoxystrobin-based products.	WHO Class U (unlikely to present an acute hazard) - Green band. Max. 2 applications per season. For 250g /litre (concentrate), max. dose is 0.4L per acre. PHI: 14 days.
			Use difenoconazole-based products mixed with other fungicides.	WHO Class II (moderately hazardous) - Yellow band. 200ml per acre (for a 250g stock). PHI: 14 days.

Kenya

CREATED/UPDATED: September 2023

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Alternaria leaf spot on lettuce

Alternaria brassicae and *A. brassicicola*

	Prevention	Monitoring	Direct Control	Direct Control	Restrictions
Lettuce leaf showing Alternaria symptoms (Sandra Jensen, Cornell University, Bugwood.org)  Alternaria leaf spots on lettuce (Sandra Jensen, Cornell University, Bugwood.org)	• This is a seed borne disease plant certified disease free seeds • Transplant only healthy seedlings. • Mulch plants with straw soon after transplanting to reduce spread of spores already in the soil through rain splash • Control weeds of all kinds including volunteer crops to reduce sources of inoculum • Practise crop rotation after two seasons of lettuce with crops such as sorghum, millet or cassava • Bury old plant debris shortly after harvest to reduce inoculum level available in the next season. • Seeds can be treated with a fungicides before sowing to prevent the disease (see details of fungicides in the yellow column)	• Monitor fields weekly looking for dark brown to black lesions with yellow halos on leaf surfaces • Look for wilting and leaf drop • Spots affect quality of leaves, therefore early control action is required when first observed • Apply direct control as soon as first symptoms are observed to reduce spread of the disease OR use a protectant fungicide as a preventative measure	• When the disease is at a low incidence, rogue out all infected plants as they are seen and destroy by burning • Collect fallen leaves regularly in the season and destroy by burning to reduce availability of spores	• When using a pesticide or botanical, always wear protective clothing and follow the instructions on the product label • Do not use chemicals with the same mode of action year after year as this can lead to resistance • Always consult the most recent list of registered pesticides of MOFA, Ghana • Apply Mancozeb (e.g. Agrithane 80WP, Benco 80WP, Damazeb 80WP Dizcozeb 80WP) at a rate of 75g/15L of water. Multi-site broad spectrum fungicide, FRAC group: M3 • Apply Captan (Merpan 50WP) at 75-120g/15L of water. Multi-site broad spectrum fungicide, FRAC group:M 4 • Treat seeds with Mancozeb at 3-4g/kg of seed before sowing. FRAC code: M3 • Treat seeds with Folpet (Folpan 50 WP) at 3-4g/kg of seed. FRAC code: M4	• WHO Class U - Unlikely to present acute hazard in normal use. Apply 3 times at 7-10 day intervals. REI-24hrs; PHI-14 days. Very toxic to aquatic organisms - do not apply near waterways • WHO Class U - Unlikely to present acute hazard in normal use. Apply 3 times in a season at 7-10 day intervals. REI-24hrs; PHI-14 days. Very toxic to aquatic organisms - do not apply near waterways • WHO Class U (Unlikely to present any hazard under normal use) • WHO Class U (Unlikely to present any hazard under normal use)



Ghana

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Alternaria on Brassica

Recognize the problem

Small dark dots first develop on leaves and later enlarge into circular spots. Later, dark green fungus cells (spores) grow on the spots. This causes circular rings to appear in the spots. Old leaf spots tear off and when the dry tissue becomes paper like it falls out creating a hole. The leaves dry up during severe attack.

Background

The disease is usually severe under humid hot conditions. The fungus is seed-borne and is mainly spread through water, wind and infected farm tools. The fungus does not survive in the soil. The disease is most common under hot or warm weather.

Management

Non-chemical control

- Planting clean seeds or seedlings is recommended.
- Avoid overhead irrigation especially during head formation.
- Farm hygiene should be practised e.g. cleaning farm tools with bleach before using them.
- Brassica weeds should be controlled to avoid the disease surviving there.
- Plants should be planted using proper spacing to avoid disease spread.
- Rotating with non-brassica crops for at least two years can control the disease.

Chemical control

- Spray with Famoxadone +cymoxanil and copper oxychloride.

Leaf damage caused by *Alternaria brassicicola*. (Photo by J.P. Tewari)



When using a pesticide, always wear protective clothing and follow the instructions on the product label, such as dosage, timing of application, and pre-harvest interval.

Scientific name(s) > *Alternaria brassicicola*

The recommendations in this factsheet are relevant to: many countries where this pest or problem is present



Authors: This factsheet is based on information written for "The Brassica Farming Handbook", first published by KENGAP Horticulture 2011.
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Edited by Plantwise.

Bacterial soft rot on cabbage

Erwinia carotovora subsp. *carotovora*



Bacterial soft rot on cabbage
(Photo: Bernard Sakwa,
MoALF, Kenya)

Prevention	Monitoring	Direct Control	Direct Control	Restrictions
• Avoid water logging by planting on raised beds or ridges. • Rotate with leguminous crops such as beans and cereal crops such as maize for three seasons to avoid build-up of the disease inoculum • Avoid injury of crop while carrying out field operations as these will be form entry points of the disease • Avoid overhead irrigation to minimize soils splash that spreads disease • Avoid harvesting during warm and moist weather	• Additional relevant crops: turnip, rutubaga, broccoli and cauliflower. • Monitor the crop daily by looking for: • Water soaked lesions which spread rapidly over and into the head, becoming soft, mushy and dark in colour emitting an offensive smell • Take action after one plant is affected	• Harvest the healthy crop first and store in a cool, airy and disinfected store • Remove and destroy by ploughing-in all diseased debris • Uproot and carry away infected plants in a container to avoid spilling the affected soil. • Apply wood ash in the affected holes after uprooting plants	• When using pesticides, wear protective gear and follow the instructions on the product label and safely dispose empty containers • Always consult recent list of registered pesticides (PCPB) • Disinfect all farm tools by dipping in 10% jik solution (3.85% sodium hypochlorite) at 50ml per litre of water for at least 5 minutes • Spray with copper based fungicides e.g. Copper oxychloride (Amicop 50WP and Cobox 50WP at 60gms/20L of water) and Copper Hydroxide (Cuprocaffaro Micro 37.5 WG and Kocide Dfat 50gms/20L of water) to minimize bacterial infection.	• WHO II (Moderately hazardous) PHI 7 days



Kenya

CREATED/UPDATED: September, 2015

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Bacterial Soft Rot on Carrot - Kenya

Pectobacterium carotovorum (previously called *Erwinia carotovora*)



Soft, watery, slimy rot on the taproot, which is grey to brown (Photo: Oregon State University)



Severe pitting of carrot roots observed after the roots are harvested and washed (Photo: Lindsey du Toit, Washington State University)



Rapid decay that consumes the core of the carrot, often leaving the epidermis intact (Photo: www.extension.umn.edu)

Prevention	Monitoring	Direct Control	Direct Control	Restrictions
- Plant carrots in well-drained soils, and water twice a week. - Rotate with legumes, soybeans and fodder grasses - Avoid injuring carrots when weeding or harvesting to prevent entry of pathogen - Control vectors that can wound carrots e.g. cabbage maggot - see yellow section. - Start working from the healthy field towards the infected area - Disinfect all farm tools in 10% jik solution (50 mls in 1 litre) of water for 5 minutes - Avoid harvesting carrots during wet conditions	- Look out for: - Soft, watery, slimy rot on the taproot, which is grey to brown with a foul smell - Rapid decay that consumes the core of the carrot, often leaving the epidermis intact - General yellowing, wilting and collapse of the foliage above ground - Severe pitting of carrot roots observed after the roots are harvested and washed - Take action as soon as yellowing and wilting symptoms are seen on one plant	Remove and burn infected plants	- Bacterial diseases have no known cure hence avoid the introducing the pathogen. Use of copper-based fungicides only suppress the pathogen. - When using a pesticide always wear protective equipment. - Follow the instructions on the product label, such as dosage, timing of application, max. number of sprays, restricted re-entry interval. - Always consult recent list of registered pesticides (PCPB). - Spray with copper-based fungicide e.g. copper oxychloride (Amicop 50WP or Cobox 50WP at 60g per 20L water). Repeat spray after 2 weeks. Maximum 2 sprays/season. - For control of cabbage maggot, drench with Azadirachtin-based products e.g. Nimbecidine EC or Achook EC at 50mls/20L of water	WHO Class II (Moderately Hazardous), PHI 7days, REI 12hrs. Toxic to aquatic organisms. WHO Class U (unlikely to cause any effects if used as required)



Kenya

CREATED/UPDATED: November 2016

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Soft rot on vegetables

Dickeya spp., *Erwinia* spp., *Pseudomonas* spp., *Pectobacterium* spp., *Bacillus* spp., *Clostridium* spp.



Soft rot of cabbage (Nilanka Herath, Department of Agriculture, Sri Lanka)



Tuber infection of potato (Public Domain - ex. SASA ©Crown Copyright)



Soft rot on carrot (Gerald Holmes, California Polytechnic State University at San Luis Obispo, Bugwood.org)

Prevention

- Locate nurseries away from production fields and transplant to well drained soils.
- Control insect pests which can cause damage (e.g. thrips, leaf hoppers, cutworms), providing entry points for the bacteria. Practice tillage before planting to expose insect pupae to natural enemies to reduce populations.
- Ensure that plants are spaced wide enough so that water can easily evaporate off the leaves.
- Fields should not be worked while plants are wet to avoid spreading the disease.
- Maintain optimal levels of fertilization to grow strong crops which are resilient to bacterial soft rot.
- Avoid overhead irrigation, particularly in the evening when it's cooler, to prevent leaves being wet for a long period of time.
- Use ground irrigation if required, starting from non-infected parts of the field to prevent bacterial disease spread from infected areas.
- Avoid injuries to plants when working to prevent bacteria from being able to easily enter the plant.
- Clean farming tools and machinery with 10% bleach or 70% alcohol before working in different areas of the field to prevent disease spread.
- Rotate with less susceptible crops such as maize, sorghum, snap beans and beets for 2-3 years after harvest.

Monitoring

- Relevant crops: potato, carrot, tomato, cucumbers, melons, squash, pumpkins, cabbage, cauliflower, bok choy.
- Monitor field looking for small to large water-soaked lesions on the leaves.
- Affected leaf tissues turn brown, becomes soft and mushy and has a foul smell. Leaves may yellow and wilt.
- Look for decaying leaves, stems and roots.
- Consider taking action as soon as the symptoms are seen.

Direct Control

- Rogue and destroy infected plants in nurseries and fields by deep burying (at least 1 m) or burning (if allowed in your area).
- Destroy plant residues and volunteer crops by deep burying or burning.
- Harvest during dry conditions to prevent spreading the disease to other crops.
- Do not store any plants displaying symptoms as the disease can spread to other plants during storage.
- Store crop in a cool, dry place which is well aerated to suppress bacterial growth.

Note: Pesticides may be available to control this pest. Please check with the Ministry of Agriculture in your country to find out which pesticides are registered in your country and the local restrictions for their use.

Bacterial leaf spot of crucifers

Pseudomonas syringae pv. *maculicola*



Cabbage leaf showing later stages of infection (Francis Nduati, Kenya)

Prevention	Monitoring	Direct Control	Direct Control	Restrictions
- Control of this bacterial disease relies on good cultural practices and other preventive methods - Avoid planting cabbage on land with continuous history of cabbage for two years - Remove and destroy by burning volunteer and leftover plants - Avoid causing injuries to plants when working on plants to reduce mechanical transmission - Avoid clustered areas where farmers plant the same crucifers - Practice rotations with corn and sorghum every two years - Set up nurseries away from major production fields and plant in well drained soils	- Monitor fields looking for small, water-soaked pinpoint lesions with yellow halos on the leaves - Look also for dark brown or purple lesions on leaves as spots merge together - Look for chlorotic and dying leaves as the disease progresses - Take action when the first symptoms are seen - Similar symptoms can be caused by fungi such as <i>Alternaria</i> spp.	Rogue and destroy infected plants in nurseries and production fields when disease density is low	No chemical for control of this disease has been reported	

Ghana

CREATED/UPDATED: May 2016

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EDITED BY: Plantwise



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Bacterial leaf spot on Brassica

Recognize the problem

Bacterial leaf spot disease rarely occurs in lowlands. It affects stems, leaves and petioles. Infections start as small, dark spots on leaves. As the disease develops, many water soaked leaf spots appear but these stay small. Older leaf spots turn brown and may or may not have a purple border around them. Symptoms on transplants may resemble downy mildew symptoms.

Background

The bacterium is found in seed and survives in the soil for over a year. It may be splashed from plant to plant by rain, wind and sprinkler irrigation. The disease is active during warm wet weather.

Management

Non-chemical control

- Plant cabbage seeds or seedlings that are free from diseases.
- Practice a crop rotation programme for 3 seasons.
- Plant resistant varieties.
- Soils should be mulched to avoid disease spread in rain and also wind.
- Farm equipment should be kept clean and disinfected to avoid disease spread.
- Changing from sprinkler to furrow irrigation limits the spread of the bacteria.
- Destroy all the infected crop material by ploughing it in to the soil, especially after harvest.

Chemical control

- Copper oxychloride can control the bacteria.

Bacterial leaf spot. (Photo by Josef Schlaghecken)



When using a pesticide, always wear protective clothing and follow the instructions on the product label, such as dosage, timing of application, and pre-harvest interval.

Scientific name(s) > *Pseudomonas syringae* pv. *maculicola*

The recommendations in this factsheet are relevant to: many countries where this pest or problem is present



Authors: This factsheet is based on information written for "The Brassica Farming Handbook", first published by KENGAP Horticulture 2011.
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Edited by Plantwise.

Ring Spot on Brassica

Recognize the problem

Ring spot affects crops such as kales, Brussels sprouts and cauliflower. All above ground parts are affected. The symptoms appear as circular brown-grey spots on the leaves, which are often surrounded by a green border with black-speckled rings. In stored cabbage, the fungus spreads deep into the leaves, requiring removal of leaves before marketing.

Background

The disease is caused by a fungus in the seed. The fungus is also found in infected crop material. Disease development is favoured by cool moist conditions. It is mainly spread by wind, water splashes and also compost made from infected crop remains.

Management

Non-chemical control

- Plant disease free transplants.
- Nurseries should be established away from infected plants.
- Practice a crop rotation programme with non-brassica for at least two years.
- Decompose all plant material immediately after harvest to enhance decay and avoid disease spread in the future.
- Brassica weeds in the farm should be removed to avoid disease spread.
- Remove neighbouring brassica plants especially at farm borders that may be infected.

Chemical control

- Use fungicides such as Azoxystrobin, Mancozeb and Copper oxychloride.

Ring spots on Brassica. (Photo by Kengap Horticulture)



When using a pesticide, always wear protective clothing and follow the instructions on the product label, such as dosage, timing of application, and pre-harvest interval.

Scientific name(s) > *Mycosphaerella brassicicola*

The recommendations in this factsheet are relevant to: many countries where this pest or problem is present



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Edited by Plantwise.

Black rot on cabbage

Xanthomonas campestris pv *campestris* Black rot on cabbage



Initial infection begins as yellow "V" shaped spot on edge of leaf (David B. Langston, University of Georgia, Bugwood.org).



Cabbage seedlings showing bacterial blight lesions (Tom Creswell, Purdue University, Bugwood.org)



Blackened vascular tissue (Gerald Holmes, Strawberry Center, Cal Poly San Luis Obispo, Bugwood.org)

Prevention	Monitoring	Direct Control	Direct Control	Restrictions
- Always use certified disease-free seed and only plant healthy cabbage seedlings. - Use disease-resistant varieties such as Riana F1 and Carina F1. - Soil solarization may also help to reduce bacterial inoculum in the soil (refer to factsheet). - If seed is not treated, seed treatment with water at 50°C can be done for 20-25 minutes (refer to factsheet). - Use drip irrigation to avoid water splashing. Use of mulch to prevent splashing of irrigation water. - Avoid water logging in the cabbage field by placing your plants on ridges. - Increase spacing between plants (50-60 cm) and between rows (60-90 cm) to maximize air flow and to reduce leaf wetness. - Rotate cabbage with non-cruciferous crops such as cereals (maize) and legumes (beans or groundnuts) for a minimum of three years. - Avoid using compost from brassic crops in cabbage fields.	- Look for V-shaped lesions that start from leaf edge going inside the leaf. - Diseased areas start as yellow and become brown over time. - Affected leaf veins turn dark brown to black. - The V-shaped lesions enlarge to levels where they cover the entire leaf which wilts and dries. - Infected seedlings may be stunted. - Disease thrives in wet and humid conditions. - Check for symptoms every week. - Take action as soon as you notice an infected plant, as bacteria multiply quickly and may cover the whole field in a very short period of time.	- Remove all infected plants and debris from the field and destroy by deep burying (30cm or more) or composting far away from the field after harvest. - When older plants get infected, harvest them quickly.	- When using a pesticide or a botanical, always wear protective clothing (PPE) and follow the instructions on product labels such as dosage and timing of application, pre harvest interval, re-entry interval and number of sprays. - Do not eat, drink or smoke while handling (mixing or applying) pesticides. - WHO class II chemicals may not be allowed in local IPM schemes. - Always consult ZEMA for updated pesticide lists. - Copper based sprays are known to help against bacterial infections. Spray copper oxychloride at the dilution rate of 40g per 16L sprayer (250g/100L of water).	WHO class II (Moderately acute hazard). Apply as soon as symptoms appear and repeat twice if needed. Pre harvest interval (PHI) is 14 days and Re-entry interval (REI) is 1 day.

Zambia

CREATED/UPDATED: 20th October 2021

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Leaf miner on Brassica

Recognize the problem

Leaf miner adults are small black flies with a bright yellow spot on their back. Females puncture leaves to feed on plant juice and lay eggs in the leaf tissues. After 2-4 days, the eggs hatch and larvae feed between the upper and lower surface of leaves making whitish lines or leaf mines. These suggest that the leaf miners are present.

Background

Leaf miners are found in old crop remains and also on crops related to brassicas. Their population is high on seedlings. The mines or tunnels can kill seedlings and in older plants, allow fungal diseases to enter the leaves. The leaves are rendered unmarketable. Leaf miners are attracted to the colour yellow.

Management

Non-chemical control

- Conserve natural enemies especially parasitoids by avoiding spraying pesticides when there are enough natural enemies to reduce the number of caterpillars in the field.
- Where possible, avoid planting next to infested fields, especially those near harvested crops.
- Plough to expose the miners so they burn in the sun and get eaten by natural enemies.
- Plant seedlings free of leaf miners.
- Practice field hygiene e.g. handpicking and destroying mined leaves, burning all infected plant parts and destroying hosts such as old crop debris.
- Use yellow sticky traps or yellow basins filled with water to attract adults.

Chemical control

- Effective insecticides to use include Abamectin, Bifenthrin and Methoxyfenozide.

Leaf mine on Brassica leaf. (Photo by Kengap Horticulture)



When using a pesticide, always wear protective clothing and follow the instructions on the product label, such as dosage, timing of application, and pre-harvest interval.

Scientific name(s) > *Liriomyza* spp.

The recommendations in this factsheet are relevant to: many countries where this pest or problem is present



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Edited by Plantwise.

Chinese kale - Downy mildew

Peronospora brassicae

	Prevention	Monitoring	Direct Control	Direct Control	Restrictions
 Symptom of Downy Mildew (www.bugguide.net)	• Ensure proper land preparation to make sure that your soil is well drained • Transplant only healthy seedlings • Provide adequate plant spacing to reduce the density of the canopy and improve aeration. • Thin plants and prune infected shoots also helps improve plant's aeration and reduce disease inoculum.	• Inspect the crop regularly • Scouting should begin when the plants are young until harvest time	• Remove infected plants and prune infected leaves/ shoots • Properly dispose of collected diseased-parts either by burning or burying them • Spray garlic bulb extract	• Note: for all pesticides usage to follow instructions on product label; Note: to avoid the development of resistance to pesticides, pesticides in the same FRAC Group should not be used continuously	
 Lesions caused by downy mildew (Gerald Holmes, Valent USA Corporation, Bugwood.org)	• Plough-under all the plant debris after harvest • Practice crop rotation with non-brassica crops • Alter planting dates to avoid rainy periods so as to avoid high disease pressure period			• Carbendazim; 2-3g/ lit; Contact action • Difenconazole (e.g. Score); Use 5ml/10lit water; Systemic action • Mancozeb Use 2-3g/1lit water; Contact action	• FRAC Group 1; WHO Class U (Unlikely to present acute hazard); Follow PHI on label • Triazoles FRAC Group 3; WHO Class II (Moderately hazardous) • FRAC Group M3; WHO Class U (Unlikely to; present acute hazard)



Cambodia

CREATED/UPDATED: September 2013

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EDITED BY: Plantwise

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Chinese kale - Downy mildew

Recognize the problem

Symptoms of downy mildew infection include small, pale yellow spots with indefinite borders on the upper leaf surface. Purplish discolouration of the upper leaf surface is seen on some hosts. A downy growth (sporangioophores) may be seen directly under the spots on the underside of the leaf, on fruits or on stems early in the morning or when foliage is wet. Older leaves usually remain attached. Affected areas on the leaf enlarge and turn brown and papery. When the disease is severe, whole leaves die.

Symptoms of downy mildew on upper leaf. (Photo by Yuan-Min Shen, Taichung District Agricultural Research and Extension Station (CC BY-NC))



Background

Conditions that favour disease development include:

- Cool, moist weather conditions
- Host weeds found in between the crops
- Crop residues in the field
- Poor plant aeration
- Overcrowding (planting in high densities)

Management

- Provide adequate plant spacing to reduce the density of the canopy and increase aeration
- Pruning of new growth also helps improve aeration
- Remove infected plants and prune infected leaves
- Avoid overhead watering. It lengthens the duration of leaf wetness and favours further development of the disease.
- Plough-under all crop residues/plant debris after harvest
- Practice crop rotation with non-brassica crops
- Weeds should be eradicated in and near seedbeds and out in the production fields
- Carbendazim (2-3g/ lit), difenocoazole (e.g. Score, 5ml/10lit water) and mancozeb (2-3g/1lit water) can be used as field sprays. Please refer to G&Y list for more details.

Symptom of downy mildew on lower leaf. (Photo by Gerald Holmes, California Polytechnic State University at San Luis Obispo (CC BY-NC))



When using a pesticide, always wear protective clothing and follow the instructions on the product label, such as dosage, timing of application, and pre-harvest interval.

Scientific name(s) > *Hyaloperonospora brassicae*

The recommendations in this factsheet are relevant to: Cambodia



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